

Partial Differential Equation Agents (PDE-Agents): A Large Language Model-Orchestrated Multi-Agent Framework for Automated Finite Element Simulations with Knowledge Graph-Augmented Reasoning

Sayan Adhikari¹, Gulshan Noorsumar¹, Øyvind Jensen¹

^aInstitute for Energy Technology (IFE), Kjeller, Norway

CRediT authorship contribution statement

Sayan Adhikari: Conceptualization, Methodology, Investigation, Software, Data Curation, Formal Analysis, Visualization, Writing – original draft, Writing – review & editing.

Gulshan Noorsumar: Conceptualization, Methodology, Investigation, Validation, Writing – review & editing.

Øyvind Jensen: Supervision, Writing – review & editing.

Acknowledgements

The authors thank the anonymous reviewers for their constructive feedback. Computational resources were provided by the Institute for Energy Technology (IFE).

Data Availability

All code, evaluation scripts, and result JSON files are available at <https://github.com/MatPro-IFE/pde-agents>. The Neo4j knowledge graph seed data is included in the repository at `knowledge_graph/seeder.py`.

*Corresponding author